

Why Industries Are Shifting FROM SILICON TO SILICON CARBIDE For Power Electronics

Silicon carbide (SiC) revolutionises power semiconductors, cutting switching losses by up to 90% and enhancing efficiency and compactness. As demand for electric mobility and sustainable energy grows, SiC drives a greener, more efficient future.

MATTHEW MOTO from IBS Electronics, **ALDO VASQUEZ, PATRICK KWOK** from DottoTech Semiconductor, and **ANDREW DIXON** from Biotech Semiconductor.

With its enhanced efficiency and durability, SiC is not just a material; it's the cornerstone of sustainable technological progress. It improves the overall performance of electronic components, positioning itself as a critical element in the future of high-power applications.

A webinar hosted by Diotec and IBS Electronics highlighted SiC's advantages over traditional silicon in applications like EV chargers and solar inverters.

SiC technology is revolutionising various sectors with its capabilities in high voltage and high frequency switching applications, which are essential for electric vehicle charging and solar inverters. Discussions on the latest SiC MOSFETs and Schottky diodes demonstrated their significant impact on efficiency and performance, making SiC a pivotal element in the future of power electronics.

Silicon carbide is more expensive than traditional silicon, yet for many manufacturers, its benefits far outweigh the cost. With an electrical breakdown field of $2.8 \times 10^6 \text{V/cm}$, SiC exhibits a ten times higher tolerance than silicon's $3.0 \times 10^5 \text{V/cm}$, enabling higher withstand voltages and efficiencies due to its low 'on' resistance. The bandgap energy of SiC is 3.26eV, three times higher than silicon's 1.12eV, which supports higher operating temperatures and boosts reliability. Additionally, SiC's electron drift velocity of $2.2 \times 10^7 \text{cm/s}$ is double that of silicon, facilitating faster switching, which can lead to size reductions and higher operating frequencies in devices. Moreover, SiC's thermal conductivity is 4.9W/cmK , three times that of silicon, leading to a lower temperature rise and superior thermal management in applications.

One might wonder if wide bandgap (WBG) materials like SiC and GaN are at a

